

Body-part-based individual feral cat identification from camera trap images using deep learning

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ABSTRACT

Feral cats (*Felis catus*) are a significant threat to Australia's native wildlife, contributing to the decline and extinction of at least 20 native mammal species through predation impacts. To improve the identification and monitoring of populations, individual identification of cats is required. This study proposes a body-part-based computer algorithmic approach that uses deep learning for individual identification from photos that can address a common challenge associated with using camera trapping, where often only a partial or obscured view of the objects of interest is presented. We investigated the discriminatory attributes of the images of four body parts of the cats: flank ('body'), back leg, front leg, and tail. We use a subset of a dataset of feral cats collected using camera traps deployed across the Glenelg and Otway regions of Victoria, Australia. Due to the skewed and imbalanced nature of images per individual in the dataset, we used a curated subset of 10 individuals, each with a relatively similar number of images, resulting in a total of 1644 images. We trained deep-learning models with a ResNet-50 backbone on these body parts individually as well as combinations of multiple body parts through feature concatenation. Results demonstrate that the body was the most discriminatory part for cat identification, with the back leg the next best part. Other parts added to the performance when they were combined. We conclude that individual cats can successfully be identified using partial body images captured using camera traps. While the body was the most distinctive part, the proposed method provides flexibility in cases where the body is obscured. This study shows that deep learning methods can meaningfully contribute to camera trap image analysis, and hence environmental conservation outcomes.

1. Introduction

Feral cats (*Felis catus*) pose a major predation threat to Australia's native wildlife, contributing to the decline and extinction of 20 mammal species, including marsupials, rodents, and bats (Burbidge et al., 2014; Doherty et al., 2016; Stobo-Wilson et al., 2022). Monitoring and managing their impact is crucial for protecting native Australian wildlife (Threat Abatement Plan for Feral Cats 2024, 2024). Traditional methods like live mark-recapture – where individual cats are physically marked (identified) and then re-trapped to estimate population size – are invasive, costly, and inefficient, prompting interest in non-invasive techniques such as using camera traps (O'Connell et al., 2011).

Using camera trapping to estimate population size leverages the fact that feral cats often have distinctive markings, allowing individual identification without physical capture. Feral cat population density is

now routinely assessed using individual identification from camera-traps. For example, studies by Rees et al. (Rees et al., 2019) and Bengsen et al. (Bengsen et al., 2011) effectively utilised camera trap images to assess feral cat abundance and population densities through spatially explicit capture-recapture methods (or mark-resight in the context of imagery) (Borchers and Efford, 2008). Automating such processes could significantly enhance feral cat monitoring efforts.

Deep learning, a branch of artificial intelligence already used for animal classification, excels at recognising patterns in images and holds the potential for identifying individual animals (Krizhevsky et al., 2017). A review of 82 studies on individual animal identification using deep learning (retrieved from the Scopus Digital Library; published between 2013 and 2023) showed that deep learning techniques, in particular convolutional neural networks (CNNs), have been widely adopted to accurately recognise and differentiate individual animals

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across various species (Akbar et al., 2024). However, most of these studies had not performed individual identification in an environment that is similar to that of feral cat monitoring. Only 18 of the 82 studies involved animals in a semi-controlled or wild environment, and only 7 of them used camera traps to collect imagery (Akbar et al., 2024). Of the 7, two studies involved individual feral cat identification in the wild. Zhou et al. (Zhou et al., 2020) tested Mask R-CNN and Faster R-CNN as identification models and used a ResNet feature extractor while Yang et al. (Yang et al., 2021) used YOLOv5.

The seven studies (Liu et al., 2019a; Liu et al., 2019b; Nepovinnikh et al., 2020; Nepovinnikh et al., 2022; Shi et al., 2023; Yang et al., 2021; Zhou et al., 2020) set in the wild using camera trap images faced similar challenges in distinguishing individuals based on unique features. First, automatic camera traps collecting images under a broad range of environmental conditions produce images varying in quality, limiting the performance of deep learning models. For example, Yang et al. (Yang et al., 2021) noted many blurry or overexposed photos in their dataset, because most feral cat images were taken at night when cats are most active. Second, automatic camera traps in the wild can produce datasets with few images for some individuals and, as also noted by Yang et al. (Yang et al., 2021), an uneven distribution of numbers of images per individual.

Importantly, the unpredictable nature of camera trap images means the deep learning models do not have a reliable single region of interest (ROI) that can be used to distinguish individuals. This is because the view, angle and posture of the animals that are captured by automatic camera traps triggered by motion cannot be controlled. The images also often contain partial or obscured views of the animals, showing only certain body parts.

Parts-based deep learning methods to identify individuals have been explored in non-animal cases. For example, Zhao et al. (Zhao et al., 2017) found that a parts-based approach effectively handled the challenges of re-identifying people in images due to body misalignment, varying illumination and background clutter. In the context of animal monitoring in the wild with camera traps, by combining multiple visible features, a parts-based method would address the limitations of current methods that depend on single, distinct features for identification, allowing resolution of animals in varying poses as well as partial or obscured view.

From the 82 studies reviewed (Akbar et al., 2024), only two studies used multiple animal body parts to identify individuals (de Silva et al., 2022; Shi et al., 2023). de Silva et al. (de Silva et al., 2022) used three views (full-body, face, and ears) of images of Asian elephants, but trained the datasets for each view separately on multiple CNNs. Similarly, Shi et al. (Shi et al., 2023) used five features (left face, front face, right face, left flank, and right flank) of photographs of captive Amur tigers and three features (front face, left flank, and right flank) from camera trap images of wild Amur leopards, which they trained separately on multiple CNNs. The researchers also combined these features to determine which combination yielded the best result. The two studies applying a parts-based approach to individual identification (de Silva et al., 2022; Shi et al., 2023) each segmented different body parts or features to achieve more robust results. While the methods and datasets varied between studies, their success suggests that a comparable approach could address the challenges of working with camera trap images for feral cat monitoring. In this study, we tested a deep learning model for individual feral cat identification from camera trap images. The main goal of this study was to investigate the viability of a parts-based concatenation model and identify which body part(s) contributes most to individual identification, to test robustness of applying a deep learning model to situations where images are incomplete or obscured.

2. Methodology

2.1. Dataset

The dataset used in this study originates from a previous project that collected imagery of feral cats using automatic camera traps deployed across the Glenelg and Otway regions of Victoria, Australia (Rees et al., 2023a; Rees et al., 2023b). We started with a subset of the dataset that consists of 15,881 images from 135 individual cats that were also used in the study by Yang et al. (Yang et al., 2021). The images were primarily captured by Reconyx Hyperfire HC600 camera traps, with a small number taken by PC900 and HF2X models. The cameras were configured to take sequences of five images whenever triggered by heat-in-motion. The images have a resolution of 1920×1080 , with daytime shots in colour and nighttime images in infrared, resulting in greyscale representations. Individual identification was carried out for all cats by Matthew Rees, as well as an independent observer for approximately half of the detections, where the location of the camera and the sequence of consecutive images aided in narrowing down the images to a specific individual (Rees et al., 2023a).

The cat with the most images had 2785 images, while the cat with the least number had 5. The distribution of images per individual is shown in Fig. 1. Fig. 2 shows the type of images within the dataset and varying image quality challenges faced. The individual cats in the dataset were already labelled, and a subset used for analysis in this study was refined through additional processing of both images and individuals.

2.2. Data processing

Before being input into the deep learning model, images were pre-processed to ensure they were in a format suitable for individual identification using different parts of the cat. Given the limited number of images per cat and imbalance in the number of images per individual, we selected a small set of features or regions of each cat for analysis; this required that we chose cats that had a sufficient number of images for each selected feature. This curation ensured that the target features were well represented across the training, validation, and testing datasets after data splitting. To address data imbalance among individuals and ensure consistent data representation, we selected a subset of 10 individual cats, each with 300 to 580 images (therefore excluding two individuals in the primary dataset that had 845 and 2785 images - to prevent the dataset from being skewed by these significantly higher image counts). For each cat, we first analysed the number of images that represented 14 features: face (front), face (left), face (right), body (right), body (left), tail (right), tail (left), front right leg, rear right leg, front left leg, rear left leg, rear (tail down), rear (tail up), and rear coat. However, this large number of features increased the risk of having too few images per feature for each individual cat. We therefore reduced this list by selecting features that represent the main parts of the cat and features that were commonly visible in the camera traps: front leg, rear leg, tail, and body. Additionally, to further simplify, we focused on features from only one side of the cat (either left or right), which reduced the feature set to four body parts per cat. Images lacking at least one visible feature from either a left or right-side view were discarded.

Table 1 illustrates the distribution of images for each individual, highlighting the split between left and right-side views and the total number of usable images after excluding those without clear features. For this study, we focused on the left side of the cat. Five cats had most usable images of their left side. The other five cats had most usable images of their right-side; images for these individuals were horizontally flipped to represent 'left-side' views. This approach ensured a uniform dataset, with the selected features being the left body, front left leg, rear left leg, and tail. Since cat tails are largely symmetrical, right-side images displaying a visible tail were also flipped and used. After images that do not have at least one clear feature were removed, for each of the

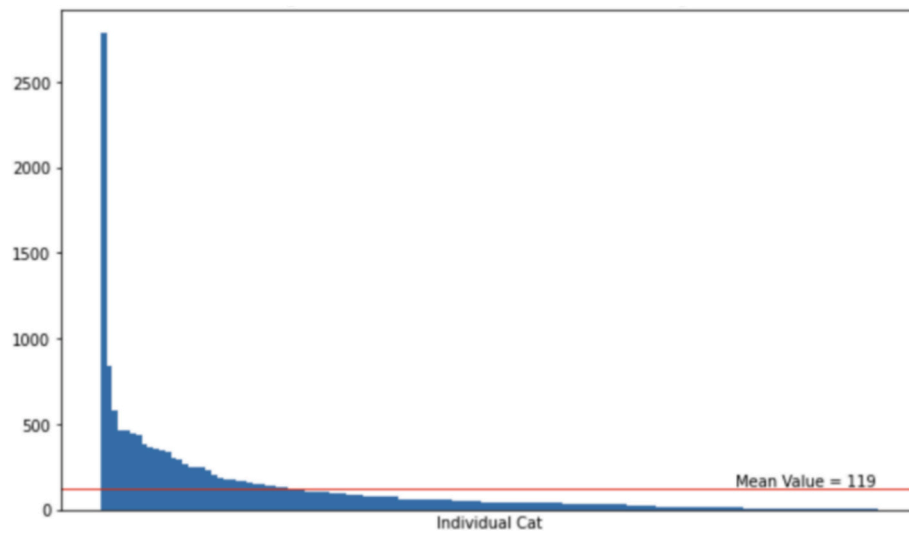


Fig. 1. Distribution of images per individual cat in the dataset (Yang et al., 2021).

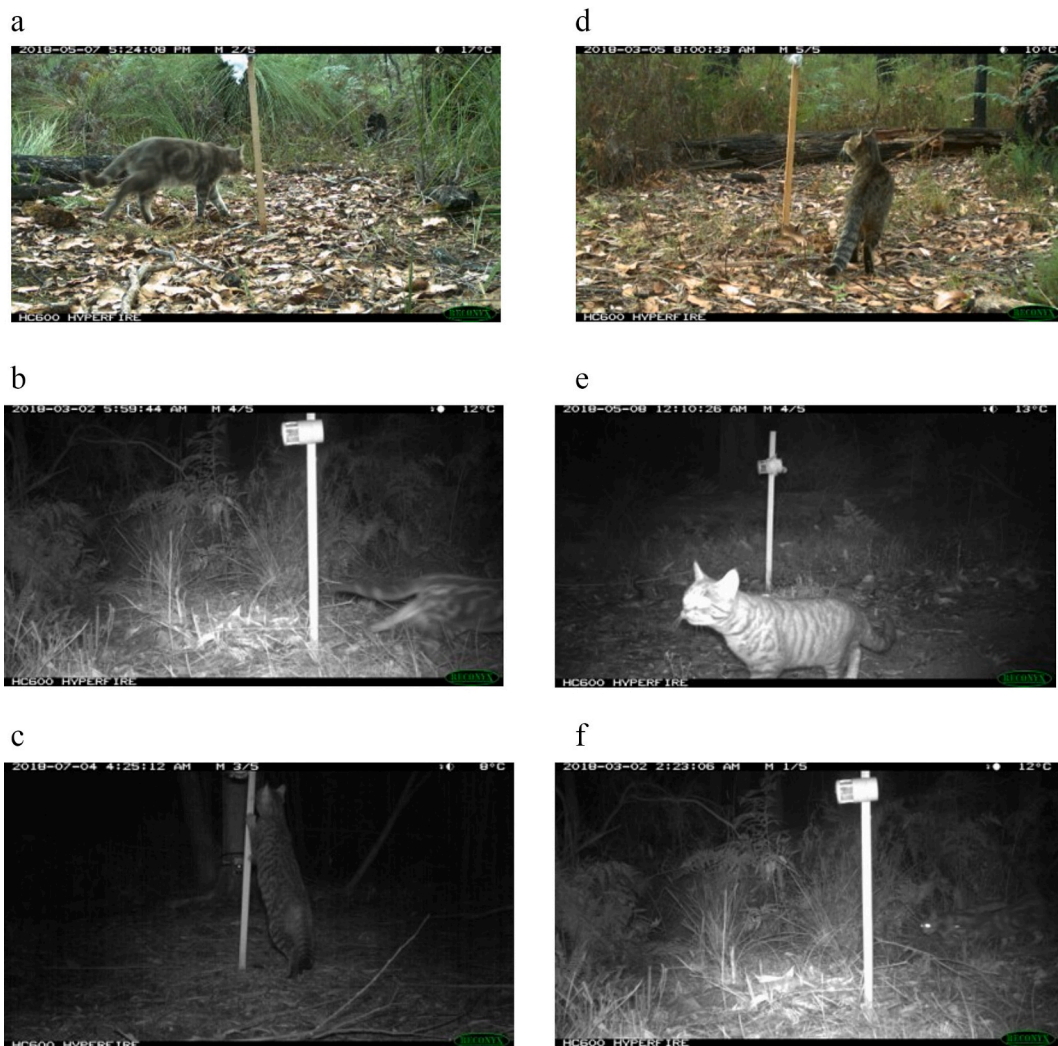


Fig. 2. Examples of cat images in the dataset. (a) Clear daytime side view of a cat, which provides an ideal example for feature identification. (b) Blurred and partial view of the cat, reducing the ability to capture a clear view of the cat or its features (c) Dark image, where the left side of the cat is visible but lacks sufficient lighting, making feature extraction difficult. (d) Image that captures only the hind view of the cat, which provides no suitable information if side views of the cat or its features are to be used. (e) Cat's left side is shown clearly, but the front and rear left legs are cut off, another example of a partial view, limiting the completeness of the information captured (f) Blurry image of two cats, meaning this image cannot be assigned to an individual.

Table 1

Total number of usable images (with at least 1 clear feature).

Cat	Initial No. of Images	Number of Usable Left Side Images	Number of Usable Right Side Images
Annya swirl ben	580	89	111
Otway tabby harriet	460	128	40
Otway tabby maximus	460	95	112
Otway tabby meg	445	132	73
Otway tabby kingfluffy	435	139	48
Otway tabby persy	385	72	79
Otway swirl chowder	368	43	46
Otway tabby squash	360	127	36
Otway swirl bluey	345	46	113
Otway tabby catsup	335	63	52
Total	4173	934	710

10 individual cats, we cropped the photos to the 4 body parts visible in the images. Images therefore contained a range of between 1 and all 4 body parts visible. Table 2 shows the number of images available for each cat representing each of the 4 selected body parts (back leg, body, front leg, and tail), providing a detailed breakdown of the usable images for the study.

2.3. Initial model training and model selection

In order to observe baseline results, the initial training included training a base ResNet50 architecture to classify the individual body parts (body, back leg, front leg, tail) of the cats into one of 10 individuals. For each body part, we split the data into 70 % training, 15 % validation, and 15 % test sets and trained the model for 10 epochs with a batch size of 32, a commonly used batch size (Krizhevsky et al., 2017). ResNet50, introduced by He et al. (He et al., 2016), is a well-established deep convolutional neural network for image classification due to its ability to efficiently capture hierarchical features through residual connections, which mitigate the vanishing gradient problem and enable deep networks to train effectively. ResNet50's architecture, with its 50 layers organised into 5 stages, allows it to learn both low and high-level features. This was particularly useful for varying image quality and feature visibility, as seen in camera trap images. The network structure is as follows:

- Stage 1: Initial 7×7 convolution followed by max-pooling.
- Stages 2–5: Residual blocks with 1×1 , 3×3 , and 1×1 convolutions. Stages 2 to 5 have 3,4,6,3 blocks, respectively.

Table 2

Number of images available for each body part per individual cat.

Cat	Back Leg	Body	Front Leg	Tail
Annya swirl ben	39	90	88	49
Otway black kingfluffy	59	125	71	49
Otway swirl bluey	45	103	35	61
Otway swirl chowder	35	39	24	23
Otway tabby catsup	41	47	44	37
Otway tabby harriet	43	104	55	62
Otway tabby maximus	51	94	86	56
Otway tabby meg	26	126	101	29
Otway tabby persy	47	64	51	54
Otway tabby squash	26	120	71	66
Total	412	912	626	486

- Final Layer: A fully connected layer with 1000 output classes, activated by a SoftMax function.

This setup was designed for the ImageNet dataset, a large-scale dataset with 1000 object categories used for pre-training.

ImageNet is crucial in the context of ResNet50, as the model was originally trained on this extensive dataset, which contains millions of images across 1000 classes. Pre-training on ImageNet enables ResNet50 to learn a broad set of features that can be fine-tuned for specific tasks, such as classifying individual cat body parts in this study. While ResNet50 was selected for its image recognition capabilities and pre-training on ImageNet, to conclude which body parts contribute most to individual identification, the focus of further experimentation was on feature concatenation to identify the best-performing combination of body parts for classification accuracy.

2.4. Feature concatenation model

After running initial tests with the ResNet50 model, we developed a Feature Concatenation model that combined features extracted from multiple parts of the cat. The model was constructed using four separate ResNet50 architectures, each responsible for extracting features from a specific body part: body, back leg, front leg, and tail. Each of these ResNet50 models excluded the top layer, specifically the fully connected layer with SoftMax activation, allowing the raw feature maps from each body part to be concatenated for further processing.

After feature extraction, the outputs of the four ResNet50 models were concatenated and passed into a single fully connected layer with SoftMax activation. This structure ensured consistency with the original architecture of ResNet50, where the concatenated features were processed in a manner similar to the final classification stage of the base model.

The architecture of the feature concatenation model can be viewed in Fig. 3 and is as follows:

- Input: Four separate ResNet50 models, each taking images from one of the selected body parts (body, back leg, front leg, and tail).
- Feature extraction: Each ResNet50 model processes the input and outputs a feature map. The fully connected layer and SoftMax activation are excluded to allow for feature extraction.
- Concatenation: The feature maps from all four ResNet50 models are concatenated into a single feature vector.
- Fully connected layer: The concatenated features are passed into a single fully connected layer with SoftMax activation, responsible for the final classification of the cat into one of 10 individual categories.

This approach allows us to leverage the unique features extracted from different parts of the cat, combining them into a single model that can make a more informed classification. By concatenating features from different body parts, the model can potentially overcome the problem of an obscured or partial view of the cat where not all features exist, and by experimenting with different combinations of missing and available body parts (removing one, two, or three parts in various tests), we're able to assess each part's relative importance to overall model accuracy.

To perform feature concatenation, the features extracted from each body part must have the same length (i.e., the number of elements in each feature vector). This requirement arises because concatenation involves combining feature vectors along a specific axis, and for this operation to be valid, the dimensions along the non-concatenation axis must match. However, the number of images for each body part varied across individual cats, creating a challenge for feature concatenation.

To resolve the mismatch in image counts for each body part, the maximum number of common images across all body parts for each cat was determined and used to ensure the feature concatenation would work. For example, the cat "Ben" had 39 images of the back leg, 90 body

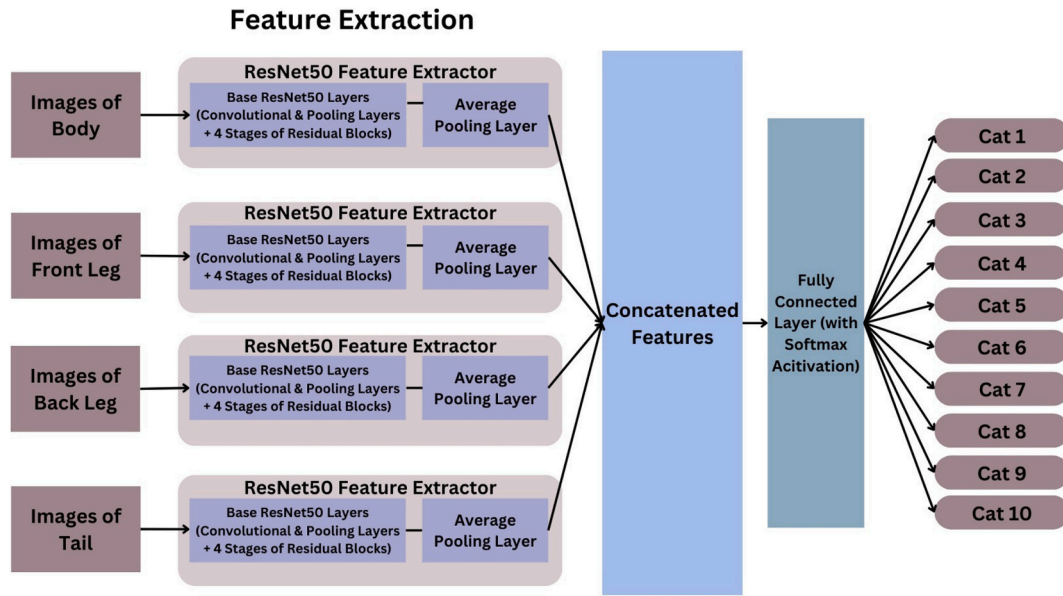


Fig. 3. Architecture of proposed feature concatenation model.

images, 88 front leg images, and 49 tail images. Since the smallest number of images for Ben was 39 (back leg images), the number representing each of the other body parts was reduced to 39 by removing additional images. This process was repeated for every feature and for each individual cat, resulting in a final dataset of 376 total images.

After ensuring the dataset was balanced across body parts, we trained the Feature Concatenation model using the combined features (body, back leg, front leg, and tail). The model was trained for 10 epochs with a batch size of 32, similar to the individual body part models. The overall results are summarised in Table 4. Further training and testing on combinations of missing and available body parts (removing 1, 2, or 3 parts in various tests) were then done to reflect situations where only partial views are available, and results were compared to assess which body part contributes the most to identification accuracy. While the Feature Concatenation model can combine all four body parts, testing with subsets of one, two, or three features demonstrates that the approach remains viable even when some features are missing and does not rely on complete or fully visible animals in each image. The key contribution of this method is to test how identification performance changes with the incremental addition of body parts, allowing us to determine which features contribute most to the complete model and to individual identification from camera trap images more broadly.

3. Results

3.1. Initial training results

The performance of the ResNet50 model for each body part is summarised in Table 3. From the results, it is evident that the body images produced the best performance, achieving the highest test accuracy (0.91) and the lowest test loss (0.26). In contrast, the other body parts (back leg, front leg, and tail) exhibited similar performance, but with lower accuracy and higher loss compared to the body. These parts had

fewer images, which likely contributed to the decreased performance. Importantly, none of the models reached 100 % training accuracy within the 10 epochs, suggesting that the model still had room for improvement and could potentially benefit from additional epochs or further tuning.

3.2. Initial feature concatenation model results

The Feature Concatenation model achieved a test accuracy of 0.92 and a test loss of 0.27, which is comparable to the performance of the ResNet50 trained on body images alone, despite using less than half the number of images. The Feature Concatenation model completed its training in 4 min, which is significantly faster than training any of the individual ResNet50 models. This improvement in speed was because the Feature Concatenation model, while architecturally similar to ResNet50, does not involve end-to-end training. The backward pass (i.e., gradient computation and weight updates) only occurs on the final fully-connected layer after feature extraction. The feature extraction portion of the ResNet50 model had frozen weights, meaning no backpropagation occurred through this part of the network, thereby reducing the computational cost and speeding up training.

3.3. Which body parts contribute the most to individual cat identification?

When excluding only one feature, the model performed best when

Table 4

Performance of feature concatenation model (all parts) (10 Epochs, Batch Size 32).

Total Images	Train Images	Val Images	Test Images	Time Taken	Test Accuracy	Test Loss
376	264	56	56	4 min	0.92	0.27

Table 3

Performance of ResNet50 on individual parts (10 Epochs, Batch Size 32).

Feature	Total Images	Train Images	Val Images	Test Images	Time Taken	Test Accuracy	Test Loss
Body	912	639	137	136	23 min	0.91	0.26
Back Leg	412	289	62	61	11 min	0.85	0.53
Front Leg	626	439	94	93	16 min	0.79	0.61
Tail	486	341	73	72	12 min	0.75	0.66

Table 5

Performance of feature concatenation model with different body parts.

Feature	Total Images	Train Images	Val Images	Test Images	Time Taken	Test Accuracy	Test Loss
a. Modelling based on all four body parts							
All included	376	264	56	56	4 min	0.92	0.27
b. Modelling based on three body parts							
Excluding Back Leg	376	264	56	56	5 min	0.92	0.28
Excluding Front Leg	376	264	56	56	4 min	0.90	0.29
Excluding Tail	376	264	56	56	5 min	0.88	0.36
Excluding Body	376	264	56	56	3 min	0.88	0.46
c. Modelling based on two body parts							
Front leg & Back Leg	376	264	56	56	3 min	0.92	0.26
Tail & Back Leg	376	264	56	56	3 min	0.90	0.29
Front Leg & Tail	376	264	56	56	4 min	0.90	0.29
Body & Back Leg	376	264	56	56	3 min	0.85	0.49
Body & Front Leg	376	264	56	56	6 min	0.83	0.46
Body & Tail	376	264	56	56	5 min	0.85	0.60
d. Modelling based on one body part							
Only Body	376	264	56	56	2 min	0.92	0.29
Only Back Leg	376	264	56	56	2 min	0.79	0.79
Only Tail	376	264	56	56	1 min	0.75	0.82
Only Front Leg	376	264	56	56	1 min	0.75	0.87

Table 6

Comparison of different models.

Model	Pros	Cons
Base ResNet50 (Single Feature)	- High performance - Accurate identification	- Cannot combine multiple features - Slow and computationally expensive
Modified ResNet50 (Single Feature)	- Faster than Base ResNet50 - Good performance with single feature	- Cannot combine multiple features - Limited to single feature usage
Feature Concatenation Model (Single Feature)	- Fast - Performs similarly to using all features	- Defeats the purpose of concatenation (since only 1 feature is used) - No backpropagation
Feature Concatenation Model (Multiple Features)	- Fast - Efficient for combining multiple body parts	- No backpropagation within feature extractor - Slightly lower performance

Table 7

Early fusion vs late fusion comparison.

Feature	Late Fusion Accuracy	Early Fusion Accuracy
a. All four parts		
All included	0.825	0.92
b. Three parts		
Excluding Back Leg	0.82	0.92
Excluding Front Leg	0.84	0.90
Excluding Tail	0.85	0.88
Excluding Body	0.80	0.88
c. Two parts		
Front leg & Back Leg	0.82	0.92
Tail & Back Leg	0.80	0.90
Front Leg & Tail	0.77	0.90
Body & Back Leg	0.88	0.85
Body & Front Leg	0.85	0.83
Body & Tail	0.83	0.85

the back leg was excluded, achieving a test accuracy of 0.92, which is comparable to the performance when all four features were used. Other exclusions, such as the front leg, tail, or body, resulted in slightly lower accuracy, with the body exclusion showing the most significant drop in performance. When using only two body parts, the model performed best when both the back leg and front leg were excluded, maintaining a test accuracy of 0.92. This performance was similar to when all four features are used. Other combinations, such as excluding the tail and back leg or excluding the tail and front leg, showed marginally lower performance. Notably, excluding the body resulted in a more significant drop in performance.

When only one body part was used, the body alone provided the best performance, achieving a test accuracy of 0.92, which is very close to the accuracy obtained when using all four body parts. The other body parts—back leg, front leg, and tail—had significantly lower performance, with accuracy around 0.75 to 0.79, showing that these parts were less informative when used in isolation. These experiments demonstrate that the body is the most critical feature for individual cat identification, providing the best performance when used alone and contributing the most when combined with other features. Excluding the body resulted in a notable drop in performance, while excluding other features (such as the back leg or front leg) only caused marginal decreases. Using two or more features improved the model's performance, but the best results were achieved when the body was included. The results are shown in Table 5.

4. Discussion

4.1. Body parts-based model for feral cat identification

A body-parts-based approach to solving the challenges of feral cat monitoring using camera traps was inspired by two previous studies that used this method to identify individual animals (de Silva et al., 2022; Shi et al., 2023). In this study, the body of the cat proved to be the most informative in distinguishing individuals. Similarly, de Silva et al. (de Silva et al., 2022) found that of the three views used (full-body, face, and ears), their Xception model trained on the ear dataset performed the best for identifying five individual Asian elephants from images, suggesting the ear region may be particularly distinctive for individual identification in Asian Elephants. Likewise, Shi et al. (Shi et al., 2023) found that the left or right flank was the most effective feature, albeit marginally,

for identifying both captive Amur tigers and wild Amur leopards, compared to other features such as different views of the face. It is likely that the ears of Asian Elephants and the flank of Amur leopards carry the most distinctive markings, and the selection of the most important feature is an important consideration in setting up the identification process.

The performance of combining features in this study did not surpass that of the ResNet50 model trained on individual body parts by any significant margin. This result supports the conclusions of Shi et al. (Shi et al., 2023), who observed that while combining features improved identification accuracy for captive Amur tigers, it did not enhance identification for wild leopards using camera trap images. In the case of leopards, the left or right body spot alone was most effective for identification, as the limited number of facial images captured by camera traps introduced noise or interference when combined with body features (Shi et al., 2023). The present study and the findings of Shi et al. (Shi et al., 2023) may be the only examples of combining features in modelling. Both studies suggest that while more features should provide additional information, in the case of camera trap images, the unpredictable pose and view of the animal often results in a limited number of images for each feature per individual. Under such conditions, these extra features may introduce interference rather than enhance performance.

From the experiments conducted, the body of feral cats is the most informative feature for individual identification. The initial results from training a stand-alone ResNet50 model on separate body parts showed that the body images produced the best performance and the lowest test loss. While the other body parts had fewer images compared to the body, the number of images for each part was still sufficient to achieve accurate results. This supports the conclusion that the body's superior performance was not solely a result of image quantity but was also due to its inherently more discriminative features.

Training the Feature Concatenation model on multiple combinations of features demonstrated that only inclusion of the body yielded similar performance as modelling using all four body parts. This suggests that the body provided the most distinguishing characteristics, allowing for accurate identification even without including the back leg, front leg, or tail. However, despite the body being the most important feature, the other body parts (such as the legs and tail) still have value. In practical scenarios where the body might be obscured in camera trap images, these additional features can provide supplementary information, contributing to more reliable identification. This result demonstrates the utility of a feature concatenation approach when working with imperfect or incomplete data from camera traps.

4.2. Value of a feature concatenation model

In the present study, the Feature Concatenation model (involving multiple features) performed similarly to an end-to-end base ResNet50 model trained solely on the body images. The key difference between these models lies in the model's speed: the Feature Concatenation model was significantly faster due to the lack of backpropagation within the ResNet50 feature extractor, as only the final fully connected layer undergoes updates during training. This makes the Feature Concatenation approach more computationally efficient. However, while the absence of backpropagation speeds up training, it may have further implications for metrics like generalisation under different conditions.

It is also important to note that the purpose of choosing a Feature Concatenation model over a model that focuses on one feature is to utilise multiple features for more robust identification, and using the Feature Concatenation model to focus on only one feature defeats the purpose of such a model. In cases where only the body was used, a modified ResNet50 optimised for speed might offer better performance and efficiency, allowing the model to take advantage of full backpropagation, while retaining a faster training time than the base ResNet50.

While the feature concatenation model does not necessarily perform better than a single feature ResNet50 model and is not functionally an end-to-end model, the Feature Concatenation model remains highly useful for ecologists in real-world scenarios where images captured by camera traps may be partially obscured. While the body may be the most important feature, having the ability to extract and concatenate features from the back leg, front leg, or tail, provides flexibility in cases where the body is not visible, ensuring that the model can still make accurate predictions.

One approach to address the challenge of unpredictable animal poses in camera trap images is to encourage specific poses using lures. Studies have shown that food and visual lures can significantly increase feral cat detection and activity rates, as well as eliciting behaviour that makes the animals more susceptible to passive monitoring techniques. For example, Paton et al. (Paton et al., 2024a) demonstrated that visual lures increased the odds of feral cat detection nearly sixfold, while Read et al. (Read et al., 2015) found that olfactory and auditory lures could trigger behavioural responses that optimise camera placements for monitoring feral predators. Lures can therefore help maximise the chances of capturing useful images for individual identification. Building on these strategies, Liu et al. (Liu et al., 2019b) introduced the Part-Pose Guided Network (PPGNet), a deep learning-based framework that uses multiple body views (e.g., fullbody, body, and limb) to normalise tiger poses captured by surveillance cameras. By aligning the animal's pose across different images, their approach improved re-identification accuracy (Liu et al., 2019b). Such pose-guided methods could complement feature concatenation approaches by reducing the variability introduced by inconsistent poses, thereby enhancing the reliability and performance of identification frameworks in challenging field conditions.

To further illustrate the trade-offs between the different models tested, Table 6 summarises the pros and cons of the most relevant models. This comparison highlights that while the Feature Concatenation model is fast and capable of handling multiple features, it lacks backpropagation, which could limit its overall performance. On the other hand, a single modified ResNet50 model might be more suitable when focusing on a standalone feature, such as the body, allowing for faster training (compared to a base Resnet50) with better optimisation through backpropagation, but cannot handle multiple features.

The Feature Concatenation method is an early fusion approach model, where features from different body parts are extracted independently, concatenated, and passed into a single classification layer. This allows the model to learn joint representations from multiple regions of the cat, even when not all parts are present. In contrast, a late fusion approach combines predictions from multiple independently trained models—one for each body part—at the decision level. Since our early fusion model combining all parts did not significantly outperform the strongest individual model (body-only, with 0.91 accuracy), we explored whether late fusion might offer an advantage by leveraging ensemble diversity. We compared its performance with a late fusion strategy based on combining predictions from independently trained ResNet50 classifiers. Across 11 combinations of body parts, early fusion outperformed late fusion in 9 cases, particularly when the body was missing or when only less informative parts were available. For instance, when only the front and back legs were used, early fusion achieved 0.92 accuracy compared to 0.82 for late fusion. Even when all four parts were included, early fusion reached 0.92, while late fusion achieved 0.825 (Table 7). These results suggest that early fusion not only provides a unified and computationally efficient model but also outperforms ensemble-style late fusion in scenarios involving partial or variable visibility. This reinforces the practical value of using feature-level integration in ecological monitoring applications where consistent full-body images may not always be available.

4.3. Limitations and recommendations for future studies

While the results of this study demonstrate the effectiveness of a

body-parts-based approach with the Feature Concatenation model, several methodological limitations should be acknowledged: First, for a more comprehensive comparison, cross-validation using different subsets of the dataset should be applied during the experiments. This approach, as highlighted by Kohavi et al. (Kohavi, 1995), ensures that the model's performance is not overly dependent on a specific training or testing set, providing a clearer assessment of robustness across varying data subsets. Cross-validation also plays an important role in supporting the validity of model performance in deep learning contexts, where traditional statistical comparisons (e.g., ANOVA) are not typically applied.

This study also focused solely on the ResNet50 architecture, leaving the potential performance of alternative CNN architectures unexplored. Other architectures (such as AlexNet, VGG, or Inception, as seen in Shi et al. (Shi et al., 2023)) may offer higher accuracy or efficiency, particularly if tailored to identify details of fine-grained body parts in challenging wildlife images. Testing these architectures could yield insights into whether certain models are more suited for multi-feature or single-feature identification in wildlife monitoring applications.

Batch size was fixed at 32; however, testing a wider range could uncover configurations better suited for this dataset, potentially improving convergence, stability, or memory efficiency (Mudigere et al., 2017). Adjusting batch sizes could also help optimise memory usage, which is particularly relevant for training complex models on large datasets of high-resolution images.

No data augmentation techniques were employed in this study, which may have limited the models' generalisation capabilities. Data augmentation typically involves applying random transformations to training images and could make the model more resilient to variation commonly encountered under field conditions, such as changes in lighting or partial occlusions (Shorten and Khoshgoftaar, 2019). Including conventional data augmentation (flipping, rotation, scaling, cropping etc.) or augmentation by generative and diffusion models in future work would help assess whether it enhances model robustness and identification accuracy under diverse real-world conditions, as well as helping with the low number of images captured for some individuals.

While our analysis evaluated overall model performance, we did not assess accuracy at the individual cat level. In ecological studies, where individuals are natural sampling units, per-cat comparisons can offer deeper insight into model consistency. Future work could incorporate a two-stage approach: calculating per-cat accuracy across models, followed by a generalised linear mixed model (GLMM) to test the effects of variables such as model type, number of images, and body parts used. This would allow for a more rigorous statistical comparison and help identify factors that influence classification performance in practical applications.

This study employed a parts-based approach primarily to evaluate the contribution of individual body parts to identification accuracy and to assess the method's viability in addressing challenges posed by partial or obscured views in camera trap images. Manual cropping was used here solely as a controlled method to isolate features and should not be interpreted as a necessary step for real-world applications. In practice, future research should focus on automating this process—for instance, by using segmentation models capable of detecting and extracting visible regions of the cat directly from full images. This would enhance scalability and usability in applied ecological monitoring.

Finally, future studies should also investigate full-image approaches, including the use of entire images as input to conventional CNNs and vision transformers. Vision transformers, in particular, offer promising potential by learning spatial relationships through patch-based attention mechanisms, which may be more adaptable to the variable quality and positioning of camera trap imagery. Exploring these techniques will provide a more comprehensive understanding of how best to approach individual identification under realistic field conditions, and may yield more robust and automated solutions for conservation efforts.

5. Conclusion

This study demonstrates that it is possible to set up a model that effectively incorporates multiple body parts for the identification of individual feral cats through camera trap imagery. The Feature Concatenation model can overcome challenges in dealing with camera trap imagery, while also exploring the comparative value of body parts in feral cat identification. Results indicate that the view of the flank of the cat (the 'body' in this study) provided the most informative features for identification. The body alone performed better than other body parts when trained on a base ResNet50 model. When trained without concatenation by the Feature Concatenation model, the same results were achieved as the Feature Concatenation model trained on all four features.

Additionally, the limited quality of night-time camera trap images likely obscured distinct patterns on other body parts, such as the tail and legs. Higher-quality daylight images or the use of white light flash, as demonstrated by Paton et al. (Paton et al., 2024a) can produce clearer images without negatively impacting feral cat behaviour or re-detection rates (Paton et al., 2024b). Incorporating such methods could enhance the visibility of markings on body parts that are currently underutilised, such as the tail and legs, allowing these features to contribute more meaningfully to identification and further improving the performance of models in future research. For species lacking distinguishing marks, however, live trapping presently remains the only viable option for individual identification.

CRedit authorship contribution statement

Rio Rifqi Syah Akbar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Matthew W. Rees:** Writing – review & editing, Visualization, Resources, Data curation. **Patricia A. Fleming:** Writing – review & editing, Supervision, Investigation, Data curation. **Ferdous Sohel:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset used in this study was originally collected as part of the study by Matthew Rees [20]. Yang et al. also used this dataset [12]. For further information about the dataset or access requests, please contact Matthew Rees (a co-author in this paper).

The code and models used in this study will be available at: <https://github.com/rio-rsa/body-part-based-feral-cat-identification/>

upon publication. This repository will include python scripts and a tutorial outlining the steps used in the analysis.

References

- Akbar, R.R.S., Fleming, P.A., Sohel, F., 2024. Individual Animal Identification through Deep Learning: A Systematic Review with Perspectives on Feral Cat Monitoring (under review; available from ResearchSquare).
- Bengsen, A., Butler, J., Masters, P., 2011. Estimating and indexing feral cat population abundances using camera traps. *Wildl. Res.* 38 (8), 732–739.
- Borchers, D.L., Efford, M.G., 2008. Spatially explicit maximum likelihood methods for capture-recapture studies. *Biometrics* 64 (2), 377–385.
- Burbidge, A., Harrison, P., Woinarski, J., 2014. The Action Plan for Australian Mammals 2012. CSIRO Publishing, Collingwood, Australia.
- de Silva, E.M.K., Kumarasinghe, P., Indrajith, K.K.D.A.K., Pushpakumara, T.V., Vimukthi, R.D.Y., de Zoysa, K., Gunawardana, K., de Silva, S., 2022. Feasibility of using convolutional neural networks for individual-identification of wild Asian elephants. *Mamm. Biol.* 102 (3), 931–941.
- Doherty, T.S., Glen, A.S., Nimmo, D.G., Ritchie, E.G., Dickman, C.R., 2016. Invasive predators and global biodiversity loss. *Proc. Natl. Acad. Sci. USA* 113 (40), 11261–11265.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* Las Vegas, NV, USA.
- Kohavi, R., 1995. A Study of cross-validation and bootstrap for accuracy estimation and model selection. In: *Proc. 14th Int. Joint Conf. Artif. Intell.* Montreal, QC, Canada.
- Krizhevsky, A., Sutskever, I., Hinton, G.E., 2017. ImageNet classification with deep convolutional neural networks. *Commun. ACM* 60 (6), 84–90.
- Liu, N., Zhao, Q., Zhang, N., Cheng, X., Zhu, J., 2019a. Pose-guided complementary features learning for Amur tiger re-identification. In: *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop.* Seoul, South Korea.
- Liu, C., Zhang, R., Guo, L., 2019b. Part-pose guided Amur tiger re-identification. In: *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop.* Seoul, South Korea.
- Mudigere, N.S.K.D., Nocedal, J., Smelyanskiy, M., Tang, P.T.P., 2017. On large-batch training for deep learning: generalization gap and sharp minim. In: *5th Int. Conf. Learn. Representations.*
- Nepovinnikh, E., Eerola, T., Kälviäinen, H., 2020. Siamese network based pelage pattern matching for ringed seal re-identification. In: *Proc. IEEE Winter Appl. Comput. Vis. Workshops.* Snowmass Village, CO, USA.
- Nepovinnikh, E., Chelak, I., Lushpanov, A., Eerola, T., Kälviäinen, H., Chirkova, O., 2022. Matching individual Ladoga ringed seals across short-term image sequences. *Mamm. Biol.* 102 (3), 957–972.
- O’Connell, A.F., Nichols, J.D., Karanth, K.U., 2011. *Camera Traps in Animal Ecology: Methods and Analyses.* Springer, New York, NY, USA.
- Paton, A.J., Brook, B.W., Buettel, J.C., 2024a. Here kitty-kitty: lure choice for predator attraction in a temperate environment. *Wildl. Res.* 51 (10).
- Paton, A.J., Buettel, J.C., Brook, B.W., 2024b. Shedding light on predator detections: evaluating the impact of camera-trap flash type for feral cat monitoring through in-field observations. *Aust. J. Zool.* 72 (4).
- Read, J.L., Bengsen, A.J., Meek, P.D., Moseby, K.E., 2015. How to snap your cat: optimum lures and their placement for attracting mammalian predators in arid Australia. *Wildl. Res.* 42 (1), 1–12.
- Rees, M.W., Pascoe, J.H., Wintle, B.A., Le Pla, M., Birnbaum, E.K., Hradsky, B.A., 2019. Unexpectedly high densities of feral cats in a rugged temperate forest. *Biol. Conserv.* 239, 108287.
- Rees, M.W., Pascoe, J.H., Le Pla, M., Robley, A., Birnbaum, E.K., Wintle, B.A., Hradsky, B.A., 2023a. Mesopredator release among invasive predators: controlling red foxes can increase feral cat density and alter their behaviour. *J. Appl. Ecol.* 60 (6), 1100–1114.
- Rees, M.W., Pascoe, J.H., Le Pla, M., Robley, A., Birnbaum, E.K., Wintle, B.A., Hradsky, B.A., 2023b. Data from: Mesopredator release among invasive predators: controlling red foxes can increase feral cat density and alter their behaviour. In: *Dryad Digital Repository.*
- Shi, C., Xu, J., Roberts, N.J., Liu, D., Jiang, G., 2023. Individual automatic detection and identification of big cats with the combination of different body parts. *Integr. Zool.* 18 (1), 157–168.
- Shorten, C., Khoshgoftaar, T.M., 2019. A survey on image data augmentation for deep learning. *J. Big Data* 6 (1), 60.
- Stobo-Wilson, A.M., et al., 2022. Counting the bodies: estimating the numbers and spatial variation of Australian reptiles, birds and mammals killed by two invasive mesopredators. *Divers. Distrib.* 28 (5), 976–991.
- Threat Abatement Plan for Feral Cats 2024, 2024. Canberra, Australia.
- Yang, Z., Sinnott, R., Ke, Q., Bailey, J., 2021. Individual feral cat identification through deep learning. In: *Proc. IEEE/ACM 8th Int. Conf. Big Data Comput., Appl. Technol. (BDCAT).* Leicester, UK.
- Zhao, H., Tian, M., Sun, S., Shao, J., Yan, J., Yi, S., Wang, X., Tang, X., 2017. Spindle net: Person re-identification with human body region guided feature decomposition and fusion. In: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* Honolulu, HI, USA.
- Zhou, J., Wang, S., Chen, Y., Sinnott, R.O., 2020. A web application for feral cat recognition through deep learning. In: *Proc. Big Data – BigData 2020.* Cham, Switzerland.